

The Department of Telecommunication Engineering is part of the Faculty of Information & Communication Technology (FICT) and is officially accredited by the Pakistan Engineering Council (PEC). The core technological focus of the department includes cutting-edge research and application in emerging fields including Blockchain, Artificial Intelligence (AI), AI-enabled telecommunication networks & systems, and 5G & beyond networks. Industrial specialization disciplines offered include RF & Microwave, Satellite Communication, Optical Fiber Communication, Wireless & Mobile Communication, and Analog & Digital Communication. The department began as an undergraduate program in 2005 and graduated its first batch in 2009, equipping approximately 50 students with telecommunication systems expertise each year. It expanded to offer an MS degree program in 2013. The program is structured to cultivate technical experts who will investigate engineering problems using modern techniques and propose effective solutions. These experts are also expected to manage projects and become effective members of engineering teams through interpersonal skills. Furthermore, they will demonstrate high moral and ethical values, a life-long learning attitude, and fulfill societal responsibilities. Admission requirements for the Bachelor of Science (BS) program include F.Sc (Pre-Engineering) or ICS with Mathematics and Physics from a recognized board with at least 60% marks, or a Diploma of Associate Engineer (DAE) in a relevant field with a minimum of 60% marks. Degree requirements involve a total of 130-145 credit hours, a total of 42 - 44 courses, and a minimum passing standing of CGPA ≥ 2.0 . The curriculum plan by semester follows a credit hour format of Theory + Lab. The 1st Semester includes HUM-163 | Functional English | Prerequisite: None | Credit Hours: 3+0, MATHP-118 | Calculus & Analytical Geometry | Prerequisite: None | Credit Hours: 3+0, HUM-101 | Islamic Studies / Ethical Behavior | Prerequisite: None | Credit Hours: 2+0, PHY-205 | Applied Physics | Prerequisite: None | Credit Hours: 3+1, CE-162L | Computer Aided Engineering Design | Prerequisite: None | Credit Hours: 0+1, CS-114 | Programming Fundamentals | Prerequisite: None | Credit Hours: 2+1, and EE-103L | Electronics Workshop | Prerequisite: None | Credit Hours: 0+1. The 2nd Semester includes MATHA-214 | Differential Equations | Prerequisite: Calculus & Analytical Geometry | Credit Hours: 3+0, HUM-268 | Communication Skills | Prerequisite: None | Credit Hours: 3+0, EE-102 | Digital Logic Design | Prerequisite: None | Credit Hours: 3+1, EE-203 | Circuit Analysis | Prerequisite: None | Credit Hours: 3+1, and CS-212 | Object Oriented Programming | Prerequisite: Programming Fundamentals | Credit Hours: 3+1. The 3rd Semester includes MATHA-213 | Complex Variables & Transforms | Prerequisite: Calculus & Analytical Geometry | Credit Hours: 3+0, HUM-102 | Pakistan Studies | Prerequisite: None | Credit Hours: 2+0, ELE-201 | Electrical Network Analysis | Prerequisite: Circuit Analysis | Credit Hours: 2+1, MATHP-111 | Linear Algebra | Prerequisite: None | Credit Hours: 3+0, CS-214 | Data Structures and Algorithms (IDEE-I) | Prerequisite: None | Credit Hours: 2+1, and EE-208 | Electronic Devices and Circuits | Prerequisite: Applied Physics | Credit Hours: 3+1. The 4th Semester includes TE-408 | Computer Communications & Networks | Prerequisite: None | Credit Hours: 3+1, CE-324 | Microprocessors and Microcontrollers | Prerequisite: Digital Logic Design | Credit Hours: 3+1, ELE-202 | Probability Methods in Engineering | Prerequisite: None | Credit Hours: 3+0, EE-261 | Signals & Systems | Prerequisite: Complex Variables and Transforms | Credit Hours: 3+0, and EE-105 | Electromagnetic Theory | Prerequisite: Calculus & Analytical Geometry | Credit Hours: 3+0. The 5th Semester includes TE-433 | Wave Propagation & Antennas | Prerequisite: Electromagnetic Theory | Credit Hours: 3+1, TE-314 | Communication Systems | Prerequisite: None | Credit Hours: 3+1, HUM-264 | Technical Report Writing | Prerequisite: None | Credit Hours: 3+0, TE-412 | Digital Signal Processing | Prerequisite: Signals

and Systems | Credit Hours: 3+1, and EE-241 | Control Systems | Prerequisite: Linear Algebra | Credit Hours: 2+1. The 6th Semester includes TE-442 | Optical Fiber Communications | Prerequisite: None | Credit Hours: 3+1, TBA | IDEE-II | Prerequisite: None | Credit Hours: 2+1, TE-313 | Digital Communications | Prerequisite: Probability Methods in Engineering | Credit Hours: 3+1, MGMT-202 | Entrepreneurship & Leadership | Prerequisite: None | Credit Hours: 3+0, and TE-415 | Transmission and Switching Systems | Prerequisite: None | Credit Hours: 3+1. The 7th Semester includes TE-432 | Microwave Engineering | Prerequisite: Wave Propagation & Antennas | Credit Hours: 3+1, TE-425 | Emerging Wireless Technologies & RF Planning | Prerequisite: None | Credit Hours: 3+0, TE-422 | MBC I (Telecom Traffic Engineering) | Prerequisite: Probability Methods in Engineering | Credit Hours: 3+0, ECON-105 | Engineering Economics | Prerequisite: None | Credit Hours: 3+0, and TE-452L | Final Year Project-I | Prerequisite: None | Credit Hours: 0+3. The 8th Semester includes MGMT-304 | Organizational Behavior or Sociology | Prerequisite: None | Credit Hours: 3+0, TE-423 | MBC II (Telecom Policies & Regulation) | Prerequisite: None | Credit Hours: 3+0, TE-441 / TE-409 | MBC III (Satellite Comm. or Network Security) | Prerequisite: None | Credit Hours: 3+0, HUM-302 | Professional Practices | Prerequisite: None | Credit Hours: 3+0, and TE-453L | Final Year Project-II | Prerequisite: None | Credit Hours: 0+3.